



DEPARTMENT OF MATHEMATICS

Academic Year 2025-2026 (Even Semester)

Date: 17 April 2026	CIE-I	Max Marks: 10+50
Time: 2:00 PM to 4:00 PM	UG	Duration: 120 minutes
Semester: II, B. E. (BT, CD, CI, CS, CY)		
Course Title: Number Theory Vector Calculus and Computational Methods		Course Code: MA221TC

- Answer all the questions.
- Answer the quiz within the first two pages of the answer booklet.
- Marks will not be awarded for direct answers without supporting steps.
- Write answers up to four decimal places.
- First year Mathematics Handbook for B.E. program is allowed.

PART-A

Q. Nos.	Quiz	M	CO	BT								
1	If $f(x) = (5x + 1)(1 - 7x)(-1 + 11x)$ and $h = 2$, then $\Delta^3 f(x) = \underline{\hspace{2cm}}$ and $\Delta^5 f(x) = \underline{\hspace{2cm}}$.	2	1	1								
2	For the following data <table><tr><td>x</td><td>-2</td><td>0</td><td>1</td></tr><tr><td>$f(x)$</td><td>-2</td><td>2</td><td>1</td></tr></table> Find x such that $f(x) = 0$	x	-2	0	1	$f(x)$	-2	2	1	2	2	2
x	-2	0	1									
$f(x)$	-2	2	1									
3	If $\gcd(6409, 42823) = g$, then for what value of x other than ± 42823 , $\gcd(6409, x) = g$.	2	2	2								
4	If p is a prime, then verify that $p + 2$ is a prime or not with examples.	2	1	1								
5	Obtain the number of divisors of 999.	2	1	1								

PART-B

Q. Nos.	TEST	M	CO	BT														
1	The following data defines the sea-level concentration of dissolved oxygen for fresh water as a function of temperature: <table><tr><td>$T, ^\circ\text{C}$</td><td>0</td><td>8</td><td>16</td><td>24</td><td>32</td><td>40</td></tr><tr><td>$O, \text{mg/L}$</td><td>14.621</td><td>11.843</td><td>9.870</td><td>8.418</td><td>7.305</td><td>6.413</td></tr></table> Using suitable interpolation estimate $O(27)$ and $O(7)$. Also, obtain $\Delta^2(24)$.	$T, ^\circ\text{C}$	0	8	16	24	32	40	$O, \text{mg/L}$	14.621	11.843	9.870	8.418	7.305	6.413	10	1	2
$T, ^\circ\text{C}$	0	8	16	24	32	40												
$O, \text{mg/L}$	14.621	11.843	9.870	8.418	7.305	6.413												
2a	For the following data, use suitable interpolation to find the interpolating polynomial $f(x)$ and hence evaluate $f(4)$. <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>5</td></tr><tr><td>$f(x)$</td><td>3</td><td>6</td><td>19</td><td>99</td></tr></table>	x	1	2	3	5	$f(x)$	3	6	19	99	6	2	2				
x	1	2	3	5														
$f(x)$	3	6	19	99														



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2b	You have to measure the flow rate of water through a small pipe. In order to do it, you place a bucket at the pipe's outlet and measure the volume in the bucket as function of time as tabulated below. <table border="1"> <tr> <td>Time</td><td>0</td><td>3</td><td>6</td><td>9</td></tr> <tr> <td>Volume</td><td>0</td><td>1</td><td>8</td><td>16.4</td></tr> </table> Estimate the flow rate and rate of the change in the flow rate at $t = 9$s.	Time	0	3	6	9	Volume	0	1	8	16.4	4	3	3
Time	0	3	6	9										
Volume	0	1	8	16.4										
3a	The normal distribution is defined by $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$. Evaluate $\int_0^3 f(x) dx$ by using (i) Simpson's 3/8 rule (ii) Weddle's rule by dividing the interval $[0,3]$ into 6 equal parts.	6	2	2										
3b	Solve the linear congruence $34x = 60 \pmod{98}$.	4	2	2										
4a	Use the Euclidean algorithm to obtain integers x and y satisfying the following: $\gcd(12378, 3054) = 12378x + 3054y$.	5	2	2										
4b	Find the last two digits of 7777^{3333}.	5	3	3										
5	Solve the following set of simultaneous congruences: $x \equiv 5 \pmod{6}$, $x \equiv 4 \pmod{11}$, $x \equiv 3 \pmod{17}$.	10	3	3										

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
Marks Distribution	Test	10	21	19	-	-	31	19	-	-	-
	Quiz	6	4	-	-	6	4	-	-	-	-